

Auteur: Noran Ben Said
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$$f(x, y) = 8xy - 2x^2 - y^4$$

$$\frac{\partial f}{\partial x} = 8y - 4x = 0 \leftrightarrow 8y = 4x \Rightarrow x = 2y$$

$$\frac{\partial f}{\partial y} = 8x - 4y^3 = 0$$

$$\Rightarrow 8(2y) - 4y^3 = 0$$

$$= 16y - 4y^3 = 4y(4 - y^2)$$

$$\Rightarrow 4 - y^2 = 0 \Rightarrow y^2 = \pm 2$$

Punten: $(0, 0), (4, 2), (-4, -2)$

$$H(x, y) = \begin{vmatrix} -4 & 8 \\ 8 & -12y^2 \end{vmatrix} \Rightarrow 48y^2 - 64$$

$$H(0, 0) = -64 \Rightarrow \text{zadelpunt}$$

$$H(4, 2) = 128 \Rightarrow \frac{\partial^2 f}{\partial x^2}(4, 2) = -4 < 0 \Rightarrow \text{Maximum}$$

$$H(-4, -2) = 128 \Rightarrow \frac{\partial^2 f}{\partial x^2}(-4, -2) = -4 < 0 \Rightarrow \text{Maximum}$$

References